SIMATS ENGINEERING

ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 10%

1. An image appears too bright. Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.
2. A noisy image contains random intensity variations. Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.
3. An image appears blurred after smoothing. Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.
4. Fine details in an image are not clearly visible. Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.
5. A noisy image must be preprocessed before further analysis. Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.
6. An object is clearly brighter than the background. Explain and justify the segmentation technique used, including its working and effectiveness.
7. You need to separate foreground and background based on intensity. Describe and apply an appropriate segmentation method, providing a clear explanation of its process.
8. An image requires identification of object edges. Explain and justify a suitable edge detection method, describing its working and effectiveness.
9. Detected edges are noisy and fragmented. Describe and justify a method to refine edges and improve boundary detection, explaining its role in segmentation.
10. An application requires precise object boundary extraction for shape analysis. *Explain and justify a suitable segmentation and boundary detection* approach, describing its working and effectiveness.

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CO_2 - AT_2 - Class test

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Q1.

when an image appears too bright, the intensity values are concentrated at the higher grey levels, making details in bright region difficult to distinguish

log transformation modifies the pixel intensity using

$$s = c \log(1+r)$$

r = input pixel intensity

s = output pixel intensity

c = constant used for scaling

Application

1. Read the bright input image
2. Normalize the pixel intensity value
3. Apply the log transformation to every pixel
4. Scale the transformed value back to the range 0-255

Expected output

* Reduces excessive brightness
* enhances details in bright regions
* Produces a clearer & better-contrasted image.

Q2. A noisy image contains random intensity variation. & justify the most suitable smoothing filter, including it's working mechanism an impact on noise reduction.

Most suitable smoothing filter : median filter

A noisy image contains random intensity variations that reduce image quality. The median filter is the most suitable smoothing filter, especially for salt - & - pepper noise

working mechanism

1. Select a small window around each pixel

2. Arrange all pixel values within the window in ascending order

3. Replace the center pixel with the median value.

Impact on noise reduction

* effectively removes random impluse noise.

* preserves image edges better than averaging filters

* Reduces unwanted intensity variations without causing significant blurring

An image appears blurred after smoothing. Explain & justify a suitable sharpening method to restore details, including it's working principle & expected results.

Suitable method: Laplacian Sharpening

Laplacian sharpening is a second-order derivate technique that enhances edges and fine details by highlighting rapid intensity changes

Formula:

$$\nabla^2 f(x,y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

Sharpened image:

$$g(x,y) = f(x,y) - \nabla^2 f(x,y)$$

Working

1. Apply the Laplacian operator
2. Detect rapid intensity changes
3. Subtract the Laplacian image from the original
4. Edge details becomes sharper

Advantage

* Restores blurred edges
* enhances fine details

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Q4. Fine details in an image are not clearly visible. Desc... Justify the filtering technique used to enhance details, explaining it's effect components.

Suitable technique : High-pass filter (HPF)

A high pass filter enhances edges and fine details by allowing high-frequency components to pass while suppressing low components.

Formula:

$$g(x,y) = f(x,y) - LPF\,(f(x,y))$$

Working

1. Remove low-frequency background
2. preserve high-frequency information
3. increases edge contrast

Advantage :

* improve edge visibility
* Highlights texture
* Increases image sharpness

Fine details becomes more visible, edge are enhanced and the image appears sharper

→ A noisy image must be preprocessed before further analysis. Explain and apply a suitable spatial filtering method

Suitable method : median filter

The median filter is an non-linear spatial filter that replaces the center pixel with the median value of it's neighboring pixels.

formula : $g(x,y) = \text{median}\{f(s,t)\}$

Working principle

1. Select a window (3x3) (5x5)
2. Arrange pixel value in ascending order
3. Repeat for all pixels

Advantage

* Remove salt and pepper noise
* preserves edges
* produces smoother image

The median filter is preferred it removes impulse noise without blurring important edges.

Q6. An object is clearly brighter than background. Exp
and justify the segmentation technique used. 13

Suitable technique: Thresholding

Thresholding separates an object from the background based on pixel intensity.

formula: $$g(x,y) = \begin{cases} 1, & f(x,y) \geq T \\ 0, & f(x,y) < T \end{cases}$$

where T is the threshold value

working

1. Select a threshold value
2. Compare every pixel with the threshold
3. pixels above the threshold becomes object(1)

Advantages

* Simple & fast
* works well when the object is brighter
* low computational cost

The bright object is accurately separated from the background, making analysis easier

You need to separate foreground and background based on intensity. explain its process.

Suitable method: otsu's thresholding

otsu's method automatically selects the optimum threshold by maximizing the variance between foreground

formula: $\sigma_B^2 = w_0 w_1 (u_0 - u_1)^2$

$w_0 w_1$ = class probabilities

u_0, u_1 = mean intensities

working principle:

1. Calculate image histogram
2. Test every possible threshold
3. Compute between - class variance
4. Select threshold with maximum variance

Advantage

* image segmentation
* medical imaging
* character recognition

Q8. An image requires identification of object edges. Expla[illegible]
& justify a suitable edge detection method

Suitable method : Canny Edge detection

Canny edge detection is a multi-stage algorithm that detects edges accurately while reducing noise.

Working :-

1. Smooth image using gaussian filter.
2. Compute gradients using Sobel operator
3. Apply double threshold
4. perform Edge tracking by hysteresis

formula

$$G = \sqrt{G_x^2 + G_y^2}$$

Clear, thin & continuous object boundaries with minimal false edge.

Detected edges are noisy and fragmented. Describe & justify

Suitable : morphological operations

morphological operation refine detected edge using a structuring element

Dilation

$A \oplus B$

Erosion

$A \ominus B$

Advantage :

* Remove isolated noise
* Connects fragmented edges
* improves segmentation accuracy

Smooth, connected boundaries with reduces noise and better segmentation

Q10. Object boundary extraction for shape analysis. Explain & justify

Suitable Approach : Thresholding + Canny Edge detection

formula :-

$$g(x,y) = \begin{cases} 1, & f(x,y) \geq T \\ 0, & f(x,y) < T \end{cases}$$

Step 2 : Boundary detection

$$G = \sqrt{G_x^2 + G_y^2}$$

Application

* medicial imaging
* industrial quality inspection
* Object measurement

A clean segmented object with accurate, continuous boundaries suitable for sharp analysis

Code of Conduct:

I Vignesh /192525013 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Vignesh

Rubrics for Evaluation

Criteria	Weigh tage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understandin g with some misconceptio ns	Poor understandin g; major misconceptio ns
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandab le with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand